

# ADEOLU ISRAEL MESOGBORIWON

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## PROFESSIONAL SUMMARY:

Results-driven Automation Engineer with hands-on experience in PLC programming, HMI/SCADA systems, and industrial control systems. Recently completed a Diploma in Mechatronics Engineering at Institute of Industrial Technology with proven knowledge in Siemens PLC S7, automation design, and electro-pneumatic/hydraulic systems. Demonstrated success in reducing equipment downtime by 65% and improving production output by 120% through strategic automation implementation. Strong foundation in process control, instrumentation, and HSE compliance from chemical engineering background and NPDC industrial training. Seeking to leverage technical skills and proven knowledge to design, implement, and troubleshoot automation systems at your organization.

## CORE COMPETENCIES:

- PLC Programming & Configuration (Siemens S7, AB PLC)
- HMI/SCADA Development (WinCC, Simatic Manager)
- Automation System Design & Implementation
- Electro-Pneumatic & Hydraulic Control Systems
- Motor Control (VFD, Star-Delta Starters, Soft Starters)
- Process Control & Instrumentation
- Troubleshooting & Maintenance of Control Systems
- HSE Compliance & Safety System Integration
- DCS Systems & Control Room Operations
- Industrial Networking & Communication Protocols

## EDUCATION & TRAINING:

### Diploma in Mechatronics Engineering

*Institute of Industrial Technology, Isheri-North, Lagos | March 2026*

### B. Eng. Chemical Engineering

*Federal University of Petroleum Resources, Effurun-Warri | November 2019*

## PROFESSIONAL EXPERIENCE:

### Industrial Automation Technician

*Institute of Industrial Technology, Lagos | August 2025 – March 2026*

- Designed and implemented HMI and PLC-based tank level control systems within 1-week project timeline, achieving automated pump operation and eliminating overflow/dry-run conditions through precise sensor integration and control logic programming.
- Automated conveyor-based sorting systems using proximity sensors and PLC logic, increasing sorting efficiency by 40% and reducing manual intervention during project execution.
- Designed, implemented, and troubleshoot Star-Delta motor starter control circuits using ON-delay timer devices and contactors, successfully reducing starting inrush current by 30% and protecting electrical infrastructure.
- Implemented variable speed control of electric motors using VFD drives, enhancing energy efficiency by 25% and providing operational flexibility for diverse production requirements.
- Designed and executed Automatic Transfer Switch (ATS) configuration for seamless power switching between grid and generator systems using relays, ensuring uninterrupted power supply with zero transition downtime.
- Developed electro-pneumatic and hydraulic control circuits using solenoid valves, directional control valves (DCV), proximity sensors, flow control valves (FCV), and actuators to achieve precise control of linear and rotary motion systems.

- Implemented sensor-based material detection switching system using PLC control logic, DCV, FCV, solenoid valves, and 3/2-way valves, reducing manual operation requirements by 60%.
- Configured and monitored Human Machine Interface (HMI)/SCADA systems using WinCC software for real-time visualization of industrial processes, improving system monitoring efficiency and operator interaction by 45%.

## **Production Supervisor**

*Industrial Projects International Ltd, Abeokuta | July 2023 – November 2024*

- Implemented structured downtime response strategies and preventive maintenance protocols, reducing equipment downtime by 65% and saving approximately ₦2.5M monthly in lost production costs.
- Improved plant production output by 120% through real-time monitoring, process control optimization, and implementation of data-driven decision-making protocols.
- Collaborated closely with maintenance teams to troubleshoot motors, VFD drives, conveyor systems, and process equipment, achieving 95% first-time fix rate.
- Supervised shift operations, ensuring compliance with safety protocols and production targets while maintaining zero lost-time accidents record.
- Managed cross-functional team of 30 operators and technicians, providing training on equipment operation and safety procedures.

## **Production Intern**

*Nigerian Petroleum Development Company (NPDC), OML 111, Benin City | July 2018 – December 2018*

- Monitored daily production activities and process parameters through DCS-based control systems to ensure optimal plant performance and process stability.
- Supported maintenance activities including equipment inspections, shutdown preparation, and commissioning procedures following industry best practices.
- Participated in safety procedures including permit-to-work systems, HAZOP studies, hazard identification, and HSE compliance checks.
- Maintained accurate production logs, daily reports, and operational records in compliance with regulatory requirements.
- Assisted in oil and water sample analysis following laboratory procedures and quality control standards.
- Gained hands-on exposure to DCS-based process systems, control room operations, and SCADA monitoring in oil and gas environment.

## **KEY TECHNICAL PROJECTS**

### **PLC-Based Automated Tank Level Control System**

- Developed complete automation solution integrating level sensors, PLC programming, and HMI interface for industrial tank management.
- Implemented fail-safe logic to prevent overflow and dry-run conditions, ensuring continuous operation and equipment protection.

### **Smart Conveyor Sorting System**

- Designed and programmed automated sorting system using proximity sensors, pneumatic actuators, and PLC control logic.
- Reduced sorting time by 40% and improved accuracy through intelligent sensor-based material detection.

### **Industrial HMI/SCADA Visualization System**

- Configured WinCC-based SCADA system for real-time process monitoring and data acquisition.
- Created intuitive operator interfaces with alarm management, trending, and historical data logging capabilities.

## CERTIFICATIONS

- Data Analysis
- HSE Level 1, 2 & 3 – Health, Safety & Environment Certification
- Health, Safety & Basic First Aid Certification

## TECHNICAL SKILLS

- **Automation & Control:** Siemens PLC S7, Simatic Manager, AB PLC, Ladder Logic Programming, Function Block Diagrams, HMI/SCADA (WinCC), DCS Systems
- **Electrical Systems:** VFD Configuration, Motor Control (AC/DC), Star-Delta Starters, Automatic Transfer Switches, Power Distribution, Industrial Wiring
- **Pneumatic/Hydraulic:** Solenoid Valves, DCV, FCV, Actuators, Circuit Design, Troubleshooting
- **Instrumentation:** Level Sensors, Proximity Sensors, Flow Meters, Pressure Transmitters, Temperature Controllers
- **Software:** Siemens TIA Portal, WinCC SCADA, AutoCAD Mechanical and Electrical, MS Office Suite

## PROFESSIONAL ATTRIBUTES

- Excellent problem-solving skills with proven ability to troubleshoot complex automation systems
- Strong attention to detail in programming, documentation, and system testing
- Safety-conscious mindset with commitment to HSE compliance and best practices
- Effective communicator with ability to work collaboratively across multidisciplinary teams
- Quick learner with adaptability to new technologies and automation platforms
- Results-oriented with track record of delivering projects on time and within specifications